

MATHEMATICS – YEAR 9

Achievement Standard Guide



About this Achievement Standard Guide

This Achievement Standard Guide describes the expected knowledge, skills and understanding students should demonstrate for Design and Technologies in Prep.

The Guides were developed using the achievement standards of Version 9.0 of the Australian Curriculum. The achievement standards assist teachers to monitor student learning and to make judgements about student progress and achievement.

The Guides set out the aspects of the achievement standard and describe indicators at five levels of attainment. They identify possible observable characteristics that could demonstrate the level of the student's understanding.

The Achievement Standard Guides were developed to:

- support teacher judgement when assessing student learning
- support planning for learning in all aspects of the achievement standard
- support the development of clear and specific learning intentions and success criteria.

The Achievement Standard Guides can be used to:

- ensure consistency of assessment
- support moderation of student work at year, school, network and state level
- support unit and lesson planning
- ensure teachers have sufficient evidence of learning to make defensible on-balance judgements about the quality of students' work against the achievement standards.

An effective assessment program will:

- Guide and support targeted teaching and learning.
- Ensure students have opportunities to demonstrate depth and breadth of their learning in all aspects of the achievement standard.
- Provide regular feedback to students about how they can improve in their learning.
- Clarify future teaching and learning needs.

The following information supports teachers in these aspects of quality assessment practice.

Mathematics Year 9 Achievement Standard

Strand	By the end of Year 9, students:
<u>Number</u>	• recognise and use rational and irrational numbers to solve problems • extend and apply the exponent laws with positive integers to variables
Algebra	• expand binomial products and factorise monic quadratic expressions • find the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment • use mathematical modelling to solve problems involving change in financial and other applied contexts, choosing to use linear and quadratic functions • graph quadratic functions and solve monic quadratic equations with integer roots algebraically • describe the effects of variation of parameters on functions and relations, using digital tools, and make connections between their graphical and algebraic representations
<u>Measurement</u>	• apply formulas to solve problems involving the surface area and volume of right prisms and cylinders • solve problems involving ratio, similarity and scale in two-dimensional situations • determine percentage errors in measurements • apply Pythagoras' theorem and use trigonometric ratios to solve problems involving right-angled triangles • use mathematical modelling to solve practical problems involving direct proportion, ratio and scale, evaluating the model and communicating their methods and findings • express small and large numbers in scientific notation
Space	• apply the enlargement transformation to images of shapes and objects, and interpret results • design, use and test algorithms based on geometric constructions or theorems
Statistics	• compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers. • explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view
<u>Probability</u>	• determine sets of outcomes for compound events and represent these in various ways • assign probabilities to the outcomes of compound events • design and conduct experiments or simulations for combined events using digital tools.

How the Achievement Standard Guide is organised

Strand	Sub-Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		Achievement standard statement with bolded variations For example, students can: • ...	Achievement standard statement For example, students can: • ...	Achievement standard statement with bolded variations For example, students can: • ...	

Content strand and sub-strand

'Can do' statements associated with each attachment level, including possible observable characteristics as examples of what students can do

Blank spaces or merged cells indicates a specific skill that can either be achieved or not.

Merged cells indicate that attainment is possible beyond 'At Standard' but not readily discernible between 'Above' or 'Well Above'.

Number

Students recognise and use rational and irrational numbers to solve problems.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Number	With support, students recognise rational and irrational numbers to solve problems. For example, students can: - Classify numbers, from a supplied list of mixed examples, as rational or irrational using supplied definitions. - Solve some problems involving real numbers using digital tools.	Students recognise and use rational and irrational numbers to solve problems. For example, students can: - Explain that a rational number can be written as a whole number or fraction, but an irrational number can't. - Classify rational and irrational numbers from a supplied list of mixed examples containing fractions, terminating decimals, recurring decimals and square roots eg: $\frac{3}{8}$, $4\frac{5}{7}$, 0.751, -9.6, 0.3, $-\frac{17}{3}$ $5.\bar{7}$, $\sqrt{16}$, $\sqrt{20}$, - Explain that we use irrational numbers because they are more accurate than rounded off decimal approximations. - Solve problems involving real numbers using digital tools and the appropriate level of accuracy for the situation.	Students recognise and use rational and irrational numbers to solve problems and justify the level of accuracy chosen for the answer. For example, students can: - Express answers as either irrational numbers or decimal approximations when answers are square roots of numbers or when answers contain π, eg an answer of $\sqrt{20}$ can be approximated as 4.5 or 4.47 or 4.472 etc. - Eg an answer of 5π can be approximated as 15.7 or 15.71 or 15.708 etc. - Understand that using irrational numbers in applied contexts results in zero percentage error whereas using decimal approximations results in non-zero percentage error and correctly justifies whether the decimal approximation is accurate enough in the situation.	

Students extend and apply the exponent laws with positive integers to variables.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Number	With support, they extend and apply the exponent laws with positive integers to numbers. For example, students can: - Apply the exponent laws to positive integers with number bases. - Eg $2^3 \times 2^4 = 2^7$	They extend and apply the exponent laws with positive integers to variables. For example, students can: - Apply any of the exponent laws one at a time with variable bases and positive integer exponents eg - First Exponent Law $x^3 \times x^4 = x^7$ - Second Exponent Law $x^5 \div x^2 = x^3$ - Third Exponent Law $x^0 = 1$ - Fourth Exponent Law $(a^3)^4 = a^{12}$ - Fifth Exponent Law $(ab)^5 = a^5b^5$ - Sixth Exponent Law $\left(\frac{a}{b}\right)^6 = \frac{a^6}{b^6}$	They extend and apply a combination of two exponent laws with positive integers to several variables and apply the seventh index law in one step situations. For example, students can: - Apply combinations of any two different exponent laws with variable bases in the same problem. - Eg simplify $\frac{x^2 \times x^5}{x^{14}}$ $= \frac{x^7}{x^{14}}$ $= \frac{1}{x^7}$ (This example uses the First then Second Exponent Laws). - Simplify $(x^2y^3)^4$ $= (x^2)^4 \times (y^3)^4$ $= x^8y^{12}$ (This example uses the Fifth then Fourth Exponent Laws) - Use the Seventh Exponent Law with bases which are numbers or variables eg $5^{-3} = \frac{1}{5^3} = \frac{1}{125}$ - eg $x^{-6} = \frac{1}{x^6}$	They extend and apply a combination of three or more exponent laws with integers to variables, including the seventh index law. For example, students can: - Apply combinations of any three (or more) different exponent laws with variable bases in the same problem (this includes using the Seventh Exponent Law). - Eg simplify $\left(\frac{x^2 \times x^7}{x^6}\right)^4$ $= \left(\frac{x^9}{x^6}\right)^4$ $= (x^3)^4$ $= x^{12}$ (This example uses the First then Second then Fourth Exponent Laws). Simplify $\frac{(4a^2bc)^2 \times (ab^2)^2}{8b^2c^2}$ $= \frac{4^2a^4b^2c^2 \times a^2b^4}{8b^2c^2}$ $= \frac{16a^6b^6c^2}{8b^2c^2}$ $= 2a^6b^4$ (This example uses the Fourth, First, then Second Index Laws)

Algebra

Students expand binomial products and factorise monic quadratic expressions.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	With support, students expand binomial products and factorise expressions. For example, students can: - Expand and factorise linear expressions eg - Expand: $3(2x + 1)$ - Factorise: $4x - 12$ - With support, make connections between expanded and factorised expressions eg - Recognise that x^2 will be a term when expanding: $(x + 1)(x + 3)$ or can recognise that the constant is 3. - Attempts an incomplete factorisation, such as $x^2 + 2x - 15$ being factorised as something like $(x + 3)(x - 5)$.	Students expand binomial products and factorise monic quadratic expressions. For example, students can: - Expand monic quadratic expressions - eg Expands: $(x + 1)(x + 8)$ - eg Expands: $(x - 2)(x + 5)$ - Factorise monic quadratic expressions of the form: $x^2 + bx + c \text{ or}$ $x^2 + bx - c \text{ or}$ $x^2 - bx + c \text{ or}$ $x^2 - bx - c$ - where c is a whole number with no more than two pairs of factors eg Factorises: $x^2 + 9x + 8$ $x^2 + 3x - 10$ $x^2 - 3x - 10$ - and double check that their answer is correct by expanding.	Students expand non-monic binomial products and factorise monic quadratic expressions, where a number of possible solutions need to be tested. For example, students can: - Expand non-monic binomial products: Eg expands: $2(x - 2)(x + 7)$ $(2x - 3)(3x + 5)$ $(2x - 3)(-3x + 5)$ - Factorise monic quadratic expressions of the form: $x^2 + bx + c \text{ or}$ $x^2 + bx - c \text{ or}$ $x^2 - bx + c \text{ or}$ $x^2 - bx - c$ - where c is a whole number with more than two pairs of factors eg Factorises: $x^2 + 11x + 24$ $x^2 - 19x - 20$ $x^2 - x - 30$	Students expand non-monic binomial products and factorise with a common factor before factorising the resultant monic quadratic expressions. For example, students can: Factorise using a combination of common factor and monic quadratic factorising techniques eg Factorises: $2x^2 + 10x - 48$ $3x^2 - 12x - 180$

Students find the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	With support, they find the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment. For example, students can: - Use a ruler and scale on a diagram OR use Pythagoras' Theorem to determine the distance between two points. - Use a process provided by the teacher to determine the midpoint between two points. - Use a process provided by the teacher to determine the gradient of a line segment.	They find the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment. For example, students can: - When given two points with coordinates which are whole numbers: - Calculate the distance between 2 points on the Cartesian Plane (eg by constructing the right-angled triangle and using Pythagoras' Theorem). - Calculate the gradient of the line containing the 2 points. - Calculate the midpoint of a line segment. - Students can also: - Determine the gradient of a line in the form of $y = mx + c$. - Calculate gradient of everyday objects, eg ramp or slide.	They use multiple strategies to solve problems related to finding the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment. For example, students can: - Determine the gradient of a line in the form $ax + by + c = 0 \text{ or } ax + by = c \text{ or } ax = by + c \text{ or } by = ax + c$ - Calculate the missing point when given one end of the interval and the midpoint. Strategies include: - Sketching the locations of the points and using horizontal and vertical distances between the points. - Scale diagram. - Using formulas.	They use multiple strategies to solve problems related to finding the distance between 2 points on the Cartesian plane, and the gradient and midpoint of a line segment when the co-ordinates of points are fractions and equations of lines are expressed with fractions. For example, students can: - Determine the gradient of a line in the form of $\frac{x}{a} + \frac{y}{b} + c = 0$ or $\frac{x}{a} + \frac{y}{b} = c$ or $\frac{x}{a} = \frac{y}{b} + c$ or $\frac{y}{b} = \frac{x}{a} + c$ - Demonstrate a number of strategies for determining the distance between two points and the midpoint of a line segment when the co-ordinates are fractions.

Students use mathematical modelling to solve problems involving change in financial and other applied contexts, choosing to use linear and quadratic functions.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	With support, students use mathematical modelling to solve problems involving change in financial and other applied contexts, using linear and quadratic functions. For example, students can:	Students use mathematical modelling to solve problems involving change in financial and other applied contexts, choosing to use linear and quadratic functions. For example, students can:	Students use mathematical modelling to solve complex problems involving change in financial and other applied contexts, choosing to use linear and quadratic functions, by converting between different representations of these functions, undertaking detailed evaluation and effectively	Students use mathematical modelling to solve complex and unfamiliar problems involving change in financial and other applied contexts, choosing to use linear and quadratic functions, converting between different representations of these functions and undertaking detailed evaluation, refining the models and effectively

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	- With support use a mathematical modelling approach that allows them to: - Translate a real-world problem into maths. - Use maths to solve the problem. - Interpret and communicate the solution back in the real-world context with some: - justification - evaluation - When using linear or quadratic rules to model financial and other applied contexts: - Classify graphical representation of real-world data points as linear or close to linear or not close to linear. - Use a supplied linear rule for to create a table of values, when the values of the independent variable are supplied, and then create a graph with correctly scaled axes. - Classify graphical representation of real-world data points as quadratic or close to quadratic or not close to quadratic. - Use a quadratic rule to complete a table of values, when the values of the independent variable are supplied, and then create a graph with correctly scaled axes. - Evaluate whether a supplied linear or quadratic rule seems	- Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths - Using maths to solve the problem - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation (with some valid reasons and checking of answers) - When using linear or quadratic rules to model financial and other applied contexts: - Find the co-ordinate pairs for points on a supplied linear graph or a parabola. - Sketch a line of best fit by eye (ie without using a defined mathematical process or technological line of best fit tool) to help find coordinate pairs for points which are in between or “outside” the original points (for linear graphs and parabolas). - Use a supplied linear or quadratic rule to create a table of values and then draw a graph with correctly scaled axes on graph paper. - Generate a table of values that matches a linear graph or a parabolic graph. - Solve a linear algebra rule to algebraically to find a numerical value of one variable when	communicating methods and findings. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths. - Using maths to solve the problem. - Interpreting and communicating the solution back in the real-world context with: - justification - evaluation and gives detailed valid reasons. (The mathematics becomes increasingly difficult, and the evaluation and justification become more sophisticated). - When using linear or quadratic rules to model financial and other applied contexts: - Determine all other representations (table, graph, algebra rule) when given a linear or quadratic mathematical model of a real-world situation in one representation. This may include using digital tools for quadratics. - When given x and y intercepts which are integers, find the linear rule which matches these points without using software	communicating methods and findings. For example, students can: - Use a mathematical modelling approach that allows them to solve practical problems by: - Translating the problem into maths. - Using maths to solve the problem. - Interpret and communicate the solution back in the real-world context - justification, including detailed reasoning, identifying the strength of the supporting evidence. - evaluation includes checking accuracy, identification of assumptions, suggestions with valid reasoning to improve the quality of the model. (The mathematics increases in difficulty, and the evaluation and justification are focussed upon modification and refinement). - When using linear or quadratic rules to model financial and other applied contexts, evaluate the effectiveness of a model by using evidence of comparison eg: - In linear modelling, students show that the values of m and c they have chosen in their rule of the form $y = mx + c$ the are the best possible values by changing m then c in their technological tool

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	to be close to real-world data points or not.	given a numerical value of the other variable. – Evaluate whether a linear or quadratic relationship (or neither) might be suitable to model data from a real-world situation based on the shape of plotted real-world data points. – Evaluate whether a supplied linear or quadratic rule seems to be close to real-world data points or not. – Interpret and communicate mathematical solutions using language (and other representations) that match the original context of the problem.	which will find the rule for them. – When given significant points (two x intercepts and the y intercept which are integers), find the quadratic rule which matches these points without using software which will find the rule for them. – Algebraically express two real-world linear rules from a worded scenario then use their rules to determine the best deal given a specified value of the independent variable. – Evaluate the effectiveness of the model by comparing the data from the model with the real-world data and commenting on whether the data from the model seems to be reasonably close to the real-world data or whether there are significant differences.	and showing that the changed models do not fit the actual real-world data as well as the original model. • In quadratic modelling, students show that the values of a, b and c they have chosen in their rule of the form $y = ax^2 + bx + c$ are the best possible values by changing a then b then c and in their technological tool and showing that the changed models do not fit the actual real-world data as well as the original model.

Students graph quadratic functions and solve monic quadratic equations with integer roots algebraically.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	With support, they graph quadratic functions and check the solutions to monic quadratic equations with integer roots algebraically. For example, students can: • Create a graph when given a table of values.	They graph quadratic functions and solve monic quadratic equations with integer roots algebraically. For example, students can: • Solve monic quadratic equations, with zero on one side of the equals sign and integer roots, algebraically:	They use significant points (y intercept, x intercepts and turning point) to sketch quadratic functions and solve more complex monic quadratic equations with integer roots algebraically by re-arranging equations to a convenient form. For example, students can:	They use significant points (y intercepts, x intercepts and turning point) to determine quadratic algebra rules, sketch quadratic functions and solve more complex monic quadratic equations with integer roots algebraically when the equation requires algebraic manipulations

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	- Create a table of values when given a graph, and from a monic quadratic algebra rule when values of the independent variable are supplied. - Substitute supplied values of x into a quadratic equation to check if they are solutions to the equation. - Write the coordinates of the y intercept and x intercepts from a supplied graph.	- in factorised form using informal methods by experimenting with values of x that make the quadratic expression equal to zero. - in expanded form by factorising first then using informal methods (experimenting with values of x that make the quadratic expression equal to zero) eg: Solve: $x^2 + 9x + 8 = 0$ Solve: $x^2 + 3x - 10 = 0$ Solve: $x^2 - 3x - 10 = 0$ - Create a table of values from a monic quadratic rule and: - Plot the graph. - Label the y-intercept (always) and the x-intercepts and the turning point (if they are present when the table of values is generated). - Identify whether the dependent variable reaches a minimum or maximum value. - Identify the values of x which make y equal to 0 from the graph. - Confirm the x intercepts (if they exist) algebraically using factorised form and substitution to show that the chosen values of x must make $y = 0$. (See the at standard dot points three sections above for the difficulty of factorising expected).	- Solve monic quadratic equations in expanded form (with zero on one side of the equals sign and integer roots) algebraically by factorising first then stating values of x that make the quadratic expression equal to zero eg: Solve: $x^2 + 10x + 24 = 0$ Solve: $x^2 - 19x - 20 = 0$ Solve: $x^2 - x - 30 = 0$ - Solve monic quadratic equations with integer roots algebraically when the terms aren't all on the same side of the equals sign to start with by moving all the terms to the same side of the equation and using factorising techniques eg: Solve: $x^2 = -10x - 24$ Solve: $x^2 - 20 = 19x$ Solve: $30 = x^2 - x$ - Algebraically find the y intercept, x intercepts and turning point to sketch a monic quadratic function. This includes: - Using the quadratic equation solving techniques described above to find the x intercepts for a parabola. - Finding the coordinates of the turning point by finding the x value halfway between the two x intercepts and then substituting this value into the rule to find the corresponding y value.	then re-arrangement to a convenient form. For example, students can: - Solve monic quadratic equations which require manipulation through expanding brackets and/or collecting like terms and moving all the terms to one side of the equals sign before solving eg: $(x + 1)(x + 3) = 8x$ $14 = (x + 2)(x - 3)$ $(x + 1)(x + 2) = -2(2x + 5)$ - Demonstrate flexibility and accuracy in using significant points to find and justify the algebra rule for a quadratic function without using graphing software eg: - Can efficiently determine and justify at least one rule which fits two x intercepts. - Can efficiently determine and justify at least one rule which fits a given y intercept and a given x intercept. - Can efficiently determine and justify at least one rule which fits a given x intercept and turning point.

Students describe the effects of variation of parameters on functions and relations, using digital tools, and make connections between their graphical and algebraic representations

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Algebra	With support, students describe the effects of variation of parameters on functions and relations, using digital tools, and make connections between their graphical and algebraic representations For example, students can: - Match tables of values to graphs. - Use a digital tool to generate a table and a graph from a linear or quadratic algebra rule. - Create a table of values from a linear or parabolic graph.	Students describe the effects of variation of parameters on functions and relations, using digital tools, and make connections between their graphical and algebraic representations For example, students can: - Use digital tools to investigate, explain and justify with evidence, the impact of varying the numerical values in linear and quadratic functions ie: - Vary m then c in linear rules of the form, $y = mx + c$ eg: $y = 2x + 5$ $y = 6x + 5$ $y = -4x + 5$ $y = 2x + 6$ $y = 2x + 10$ $y = 2x - 7$ $y = -4x - 7$ - and try other values of m and c - Vary m then n in quadratic rules of the form, $(x + m)(x + n) = y$ eg: $(x + 3)(x + 7) = y$ $(x + 3)(x - 7) = y$ $(x - 3)(x + 7) = y$ $(x - 3)(x - 7) = y$ - and try other values of m and n	Students use variation of parameters of linear functions to predict the relative positions of linear sketch graphs and describe the effects of variation of parameters on quadratic functions in expanded form, using digital tools, to make connections between their graphical and algebraic representations For example, students can: - Use what has been learnt from experimenting with digital tools to correctly predict the impact of changing parameters on graphs of linear rules (without using graphing software), ie use the values of m and c to sketch a linear rule of the form $y = mx + c$, then sketch the position of a new graphs in relation to the original graph when m changes and when c changes (labelling the y – intercepts). - Use digital tools to investigate, explain and justify with evidence, the impact of varying the coefficients and constant of quadratic functions in expanded form, ie: - Vary a then b then c in quadratic rules of the form, $ax^2 + bx + c = y$ eg: $2x^2 + 5x + 7 = y$ $2x^2 + 5x + 10 = y$ $2x^2 + 5x - 3 = y$ $2x^2 + 6x + 7 = y$ $2x^2 + 8x + 7 = y$ $2x^2 - 3x + 7 = y$, $4x^2 + 5x + 7 = y$ $6x^2 + 5x + 7 = y$ $-3x^2 + 5x + 7 = y$ - and try other values of a, b and c	Students use variation of parameters of quadratic functions in factorised and expanded form to predict the relative positions of parabolic sketch graphs. For example, students can: - Use what has been learnt from experimenting with digital tools to correctly predict the impact of changing parameters on graphs of quadratic rules (without using graphing software), ie - Use values of m and n to sketch a quadratic rule of the form $(x + m)(x + n) = y$, then correctly sketch the position of new graphs in relation to the original graph when m changes and when n changes (labelling the x and y intercepts). - Use values of a, b and c to sketch a quadratic rule of the form $ax^2 + bx + c = y$, then correctly sketch the position of new graphs in relation to the original graph when a changes, when b changes and when c changes (labelling the y-intercepts).

Measurement

Students apply formulas to solve problems involving the surface area and volume of right prisms and cylinders.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, and given formulas, students can solve problems involving the surface area and volume of right prisms and cylinders. For example, students can: - Determine the surface area of a rectangular or triangular prism, and cylinder in real-world problems when the relevant formulas for the area of the faces are supplied. - Determine the volume of a rectangular or triangular prism and cylinder in a real-world problem when relevant formulas are supplied.	They apply formulas to solve problems involving the surface area and volume of right prisms and cylinders. For example, students can: - Solve problems which require the calculation of: - Volume of right prisms (rectangular and triangular prisms) and cylinders. - Surface area of right prisms (rectangular and triangular prisms) and cylinders through recognising the components of the object and their associated formulas. - Understand that the same types of 3D objects can have the same volume but different surface areas eg: - Two rectangular prisms can have the same volume but different surface areas. - Two cylinders can have the same volume but different surface areas. - Correctly select relevant formulae from a list where relevant and irrelevant formulae are supplied.	They apply formulas to solve problems involving the surface area and volume of right prisms and cylinders where parts of the shape are missing or where they need to work backwards to find missing dimensions. For example, students can: - Determine the surface area or volume of a prism or cylinder with a missing section (eg hole/lid) in a prism (eg brick) or cylinder when the hole connects and is perpendicular to the flat surfaces, not a curved surface. - Demonstrate that rectangular prisms and/or cylinders can have different surface areas and the same volume. - Find a missing dimension in a rectangular prism, triangular prism or cylinder using the volume and the other necessary dimensions. - Find a missing dimension using the surface area and the other necessary dimensions for a rectangular prism. - Find the height of a cylinder using the surface area and the radius.	They apply formulas to solve problems involving the surface area and volume of right prisms and cylinders to optimise the surface areas of square prisms and cylinders of a given volume, and to find multiple sets of dimensions for rectangular and triangular prisms of a given volume. For example, students can: - Independently optimise the surface area of a square prism and cylinder, given the volume, ie - Correctly finds the minimum surface area for the given volume. - Presents complete and organised evidence that their answer is correct. - Uses formulas in a digital tool (eg Excel) to increase the efficiency of their data generation. - Provides a logical and correct explanation of their process with justification of each formula that they have used. - Generates at least three sets of correct dimensions for a rectangular prism or triangular prism of a given volume, justifies that the chosen dimensions are correct and compares the surface areas with clear communication of all thinking steps.

Students solve problems involving ratio, similarity and scale in two-dimensional situations.

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Measurement	With support, students solve problems involving ratio, similarity and scale in two-dimensional situations. For example, students can: - Use a non-standard reference measurement such as one block to multiplicatively determine the length of a box that is known to be 27 blocks long if told that one block is 3.2 cm long. - Work with simple whole number scale factors like 2 or 3 to make calculations and create similar shapes. For example, draw a rectangle with side lengths twice as long as the given shape.	Students solve problems involving ratio, similarity and scale in two-dimensional situations For example, students can: - Use a non-standard measurement as a reference point, eg the height of a person to estimate the height of something higher and suggest ways to improve accuracy. - Convert between the measurement on a map or scale model, to the real-life measurement and the reverse. - Calculate the whole number or unit fraction scale factor by comparing similar shapes. - Create a similar shape using the whole number and unit fraction scale factor provided. - For example, draw a triangle with side lengths 5 times as long as the given triangle. - Draw a square with a side-length one third that of the given square.	Students solve problems involving ratio, similarity and scale in two-dimensional situations that require decimal and non-unit fraction scale factors. For example, students can: - Use fractional scale factors to create similar shapes. For example, give the dimensions of a similar triangle that has lengths which are $\frac{4}{5}$ of the lengths in the given triangle. - Apply their understanding of reciprocal fractional and decimal scale factors to return to an original shape. For example: - If given an enlargement with a scale factor of $\frac{3}{2}$, students can determine the reciprocal scale factor needed ($\frac{2}{3}$) and apply this to return to the original. - If given an enlargement with a scale factor of 1.8, students can determine the reciprocal scale factor needed ($\frac{1}{1.8}$) and apply this to return to the original.	Students solve problems involving ratio, similarity and scale in two-dimensional situations that involve additional complexity such as multiple steps or irrational scale factors for relating area and length. For example, students can: - Apply irrational scale factors in multi-step real-world situations. For example, given the scale factor and dimensions of one size of paper, determine the dimensions of other sizes of paper. For example, work from A4 to A5 or A3 to A2 etc. - Solve scale problems related to area rather than length. Eg Adjust scale for a compounding calculation. For example, can calculate the dimensions of a triangle that has a fraction of the area of the original triangle.

Students determine percentage errors in measurements.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, students determine percentage errors in measurements. For example, students can:	They determine percentage errors in measurements. For example, students can:	They use percentage errors to compare errors in different measurements.	They analyse percentage errors to determine measurements, make comparisons and justify solutions.

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	- With support, calculate percentage error. - Specify the absolute errors in measurements, eg within 1 mm or within 1 cm. - State a range for errors in measurements, eg within 1 mm or within 1 cm. - Recognise that all measurements are estimates.	Calculate the absolute and percentage errors in measurements in quantities that can be measured directly, eg time, length, mass, angle, temperature.	For example, students can: Calculate and compare percentage errors to show that for example, a 1 mm error in a 5 cm length measurement is worse than a 1 mm error in a 5 m length measurement.	For example, students can: - Use a given percentage error (eg the difference between an estimation and a measurement) to identify the original estimate or measurement. - Compare and evaluate the influence of different percentage errors upon profit margin, for example, costs and benefits of a tiling contractor working with 1%, 5% and 10% percentage errors.

Students apply Pythagoras' theorem and use trigonometric ratios to solve problems involving right-angled triangles.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, students apply Pythagoras' theorem and use trigonometric ratios to solve problems involving right-angled triangles. For example, students can: - Use Pythagoras' Theorem to confirm if a triangle is a right-angle triangle. - Determine the hypotenuse length in a right-angle triangle, given the length of the other two sides, when the diagram is supplied. - Label sides of a right-angle triangle as hypotenuse, opposite and adjacent in relation to a given angle. - Determine if they should be using sin, cos or tan ratio to solve problems. - Follow a supplied set of steps to find unknown side lengths (not the	Students apply Pythagoras' theorem and use trigonometric ratios to solve problems involving right-angled triangles. For example, students can: - Use Pythagoras' Theorem to solve straight forward 2D problems by finding unknown side lengths where only the necessary information is provided and the triangle is fairly easy to draw from the worded description. - Use similarity to explain the meanings of tan, sin and cos. - Use tan, sin or cos ratios or algebra rules to find unknown side lengths and unknown angles from a given diagram. Transposition is not expected at standard but students should be able to start with one form of a rule, eg	Students apply Pythagoras' theorem and use trigonometric ratios to solve more complex problems using right-angled triangles, where necessary and unnecessary information is provided. For example, students can: - Use Pythagoras' Theorem and trigonometry to solve more involved 2D worded problems involving Pythagoras' Theorem and trigonometry to find unknown side lengths and angles where: - Both necessary and unnecessary information is provided (including unnecessary measurements). - The right-angled triangle needs to be recognised and 'extracted' from the overall diagram/scenario.	Students apply Pythagoras' theorem and use trigonometric ratios to solve more complex problems involving right-angled triangles where necessary and unnecessary information is provided, and 3D objects are sometimes involved. For example, students can: - Use Pythagoras' Theorem and trigonometry to solve more involved 2D and 3D worded problems finding unknown side lengths and angles where: - Both necessary and unnecessary information is provided (including unnecessary measurements). - The right-angled triangle needs to be recognised and 'extracted' from the overall diagram of the 3D situation.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	hypotenuse) using Pythagoras' Theorem when the diagram is supplied. Follow a supplied set of steps to find unknown side lengths and angles using trigonometry when the diagram is supplied.	$\tan(K) \times Adj = Opp$, substitute the relevant numbers, then solve the equation using inverse operations if necessary, eg when finding the adjacent side length when the angle is 34° and the opposite side length is 6 m, the solution steps could look like: $\tan(K) \times Adj = Opp$ $\tan(34) \times Adj = 6$ $0.6745 \times Adj = 6$ $Adj = \frac{6}{0.6745}$ $Adj = 8.90 \text{ m}$ - Use trigonometry to find unknown side lengths and angles in 2D worded problems where only the necessary information is provided and the triangle is fairly easy to draw from the worded description. - Correctly select relevant formulae from a list (students are not just given the formula they need for the problem they are working on).	- Unit conversions may be required For example: - Given the heights of two poles and the distance between them, work out the minimum length of a rigid flying fox track that goes from the top of one pole to the other and how steep the track will be (ie what angle will the track make with the horizontal?) - An isosceles triangle has two equal angles of 30° and two equal side lengths of 10.4 cm. Find the length of the other side of the triangle. - Transpose the trigonometric algebra rules into a convenient form to suit the problem being solved, eg $\tan(K) \times Adj = Opp$ $Adj = \frac{Opp}{\tan(K)}$ $\tan(K) = \frac{Opp}{Adj}$	- Unit conversions may be required. For example: - Given the length, width and height of a rectangular prism container, find the length of a strengthening piece of metal that goes from one corner of the container to the other and through the centre of the container. - Given the base length and height of a square pyramid that is to be constructed from metal rods, how long will the slant edges need to be and at what angle to the base should they be fastened? - Two of the side lengths of a right-angle triangle are 9 cm and 15 cm. What could the reference angle be? Show all possible answers? Use diagrams to help explain your answer. - A 2.80 m ladder is leaning against a 6.21 m high wall. The angle between the wall and the ladder is $65^\circ 25'$. What angle would the same ladder need to make with the ground to reach 20 cm higher up the wall?

Students use mathematical modelling to solve practical problems involving direct proportion, ratio and scale, evaluating the model and communicating their methods and findings.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support students use mathematical modelling to solve simple practical problems involving direct proportion, ratio and scale, evaluating the model and	Students use mathematical modelling to solve practical problems involving direct proportion, ratio and scale, evaluating the model and communicating their methods and findings.	Students use mathematical modelling to solve practical problems involving direct proportion, ratio and scale, evaluating the model and effectively	Students use mathematical modelling to solve practical problems involving direct proportion, ratio and scale, evaluating the model and effectively communicating their methods and findings with detailed reasons and

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	communicating their methods and findings. For example, students can: • With support use a mathematical modelling approach that allows them to: – Translate a real-world problem into maths. – Use maths to solve the problem. – Interpret and communicate the solution back in the real-world context with: ▪ justification ▪ evaluation. • With support, create a scale model to demonstrate understanding of ratio, similarity and scale in two-dimensional situations and then use this model to solve scenarios related to length and area of a real-world object. For example: • accurately develop a scale model of the gymnasium and all the court markings. • Communicate findings with some justification and evaluation of the model.	For example, students can: • Use a mathematical modelling approach that allows them to solve practical problems by: – Translating the problem into maths. – Using maths to solve the problem. – Interpreting and communicating the solution back in the real-world context with: ▪ justification ▪ evaluation (with some valid reasons and checking of answers). • Create a scale model to demonstrate understanding of ratio, similarity and scale in two-dimensional situations and then use this model to solve scenarios related to length and area of a real-world object. For example: • Accurately develop a scale model (using a whole number or unit fraction scale factor) of the gymnasium and all the court markings and use this to design a miniature basketball boardgame. • Evaluate your model by confirming and justifying the dimensions of the court • Develop a model to enlarge and reduce a given shape by a whole number or unit fraction scale factor. For example:	communicating their methods and findings with detailed reasons. For example, students can: • Use a mathematical modelling approach that allows them to solve practical problems by: – Translating the problem into maths. – Using maths to solve the problem. – Interpreting and communicating the solution back in the real-world context with: ▪ justification ▪ evaluation and detailed valid reasons. • Create a scale model to demonstrate understanding of ratio, similarity and scale in two-dimensional situations and then use this model to solve scenarios related to length and area of a real-world object. For example: • Adjust the scale model to be in proportion to a different sized ball (eg a tennis ball). • Evaluate your model by confirming and justifying the dimensions of the court. • Develop a model to enlarge and reduce a given shape by rational scale factors (non-unit fractions, decimals). For example:	use a generalisation to scale all measurements up or down. For example, students can: • Use a mathematical modelling approach that allows them to solve practical problems by: – Translating the problem into maths. – Using maths to solve the problem. – Interpreting and communicating the solution back in the real-world context with: ▪ justification ▪ evaluation and detailed valid reasons. – Generate an algebra rule for keeping all the measurements in proportion if one measurement changes and use a digital tool (eg Excel) to generate the new measurements using the rule. • Create a scale model to demonstrate understanding of ratio, similarity and scale in two-dimensional situations and then use this model to solve scenarios related to length and area of a real-world object. For example: • Using your scale model of the gymnasium to algebraically determine the dimensions of a basketball court for any sized ball with proportion being maintained for all other measurements. Determine the approximate height

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		- Develop a model to enlarge or reduce a logo to fit on smaller or larger item and explain your method and evaluate whether proportion has been maintained in different sizes of the same image by confirming that the dimensions maintain proportion. - Evaluate given models of the same object and determine if proportion is maintained or not. - Develop a model to represent quantities/amounts and evaluate the model for different values. For example: - Develop a model to determine your gross pay in a casual job where the base pay is \$16/hour Monday to Friday and penalty rates of time and a half on Saturday and double time on Sunday apply.	- develop a model to enlarge or reduce a logo to fit on smaller or larger item and explain your method and evaluate by confirming that the dimensions maintain proportion for any size. - Evaluate given models of the same object and determine which aspects are disproportional and identify what needs to be corrected so that proportion can be maintained. - Develop a model to represent quantities/amounts working up and back from a given value. For example: - Use a model you have developed to determine a work pattern that would result in a total gross pay amount for a casual job where the base pay is \$16/hour Monday to Friday and penalty rates of time and a half on Saturday and double time on Sunday apply.	of the players relative to a tennis ball. - Evaluate any factors that would impact accuracy and any potential sources of error. - Evaluate given models of the same object and identify and correct disproportion in the models to ensure proportion is maintained. - Develop a mathematical model that allows you to enlarge and reduce the size of a 2D shape or 3D object and translate into real-world solutions. For example: - Develop mathematical models and apply scale factor to design situations, for example, dress sizes in which scales might be different for different aspects, eg length and width change at different rates and evaluate the model identifying potential issues with the designs. - Develop a model that can be to represent quantities/amounts working up and back from a given value. For example: - Develop a generalised model that can determine work patterns that would result in a total gross pay amount for a casual job where the base pay is \$16/hour Monday to Friday and penalty rates of time and a half on Saturday and double time on Sunday apply.

Students express small and large numbers in scientific notation.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Measurement	With support, students match small and large numbers in scientific notation and decimal notation. For example, students can: - Use a supplied process and example of converting decimal notation into scientific notation to convert similar decimal numbers into scientific notation and vice versa. - Match supplied numbers in decimal notation with supplied numbers in scientific notation and vice versa.	Students express small and large numbers in scientific notation. For example, students can: - Convert from scientific notation to decimal notation and the reverse. - Give real life examples of where scientific notation might be used to represent very large and very small numbers and explain why scientific notation is useful in these situations.	Students express small and large numbers in scientific notation and utilise scientific notation to solve real-world problems. For example, students can: - Order a list of numbers where some numbers are in decimal notation and some are in scientific notation and justify the order by converting all the numbers to decimal or scientific notation. - Convert numbers expressed in words (eg 5.3 quadrillion, eg 5.3 nanometres) into scientific notation, then use the numbers in scientific notation to solve a practical problem (calculators may be used) then convert the scientific notation answer back into words when expressing the final answer to the problem.	

Space

Students apply the enlargement transformation to images of shapes and objects, and interpret results.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	With support, they apply the enlargement transformation to images of shapes and objects, and interpret results. For example, students can: Give centre of enlargement and scale factor and show how to locate first vertex (start process and student finishing).	They apply the enlargement transformation to images of shapes and objects, and interpret results. For example, students can: - Apply the enlargement transformation to enlarge or reduce shapes and objects, using dynamic geometry software as appropriate. - Identify and explain aspects that remain the same and those that change when shapes and objects are enlarged or reduced using the enlargement transformation. - Use a specified centre of enlargement and whole number or unit fraction scale factor to transform a 2D shape or rectangular prism by hand and check that the angles have remained the same, eg make the side lengths 5 times longer or one third of the original side lengths.	They apply the enlargement transformation to images of shapes and objects, and interpret results when using decimal and non-unit fraction scale factors. For example, students can: - Use a specified centre of enlargement and fractional scale factor (eg $1\frac{1}{4}$ or $\frac{4}{5}$) to enlarge or reduce a 2D shape or rectangular or triangular prism or cylinder by hand and check that the angles have remained the same. - Use reciprocal fractional and decimal scale factors and the enlargement transformation to return an enlarged or reduced shape to its original size, eg - If given an enlargement with a scale factor of $3/2$, students can determine the reciprocal scale factor needed ($2/3$) and apply this to return to the original.	They apply the enlargement transformation to images of shapes and objects, and interpret results in situations that require multiple thinking steps and irrational scale factors for relating length to area and length to volume. For example, students can: - Use the enlargement transformation to create a series of at least 5 similar shapes (with a consistent scale factor), eg a set of simple flat drawings of 5 babushka dolls, cut out to show that they do fit inside each other. - When given two similar shapes, determine the centre of enlargement and scale factor and hence construct a third shape that is similar to the first two. - Use irrational scale factors together with the enlargement transformation to create shapes and objects that have areas or volumes that are a specified multiple or fraction of the original, eg enlarge a triangle so that its area is 5 times or three fifths of the original area. Enlarge a rectangular prism so that its volume is 5 times or three fifths of the original volume. Calculate the area/volume of the original shape/object and the new shape/object to confirm that the enlargement transformation has met the specifications.

Students design, use and test algorithms based on geometric constructions or theorems.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Space	Students follow algorithms based on geometric constructions or theorems. For example, students can: Follow a given set of ordered instructions and may be able to partially explain an algorithm to say, construct an equilateral triangle.	Students design, use and test algorithms based on geometric constructions or theorems. For example, students can: Develop a set of ordered instructions, with visual representations to model scenarios eg create sets of Pythagorean Triples, construct a right-angled triangle with set of Pythagorean Triples, or bisect angles.	Students design, use and test and refine algorithms based on geometric constructions or theorems. For example, students can: - Discuss and evaluate refinements, such as only identifying the 'primitive' Pythagorean Triples, i.e, those with a Greatest Common Factor of 1; construct a right-angled triangle with set of Pythagorean Triples; or bisect reflex angles. - Explain the reasoning behind why the model does or does not work. - Design test and use an algorithm to construct an equilateral triangle, and modify the algorithm to create other regular polygons.	

Statistics

Students compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, students compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers. For example, students can: - Construct appropriate, easily comparable graphs aligned vertically or horizontally (eg histograms and divided bar graphs with grouped data and relative frequencies) by following a process given by the teacher. - Describe the shapes of data distributions (ie symmetrical, positively skewed, negatively skewed, bimodal) when given examples of data sets with these different shapes to enable easy comparison. - Follow a supplied process to determine the mean, median and range for data sets. - Draw conclusions which match the data when prompted. - Provide some valid reasons for conclusions when prompted.	Students compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers. For example, students can: - Construct a histogram and a divided bar graph using grouped data and relative frequencies. - Compare two data sets by: - Choosing and constructing appropriate, easily comparable graphs aligned vertically or horizontally (eg histograms OR divided bar graphs with grouped data and relative frequencies) - Describing the shapes of the data distributions (ie symmetrical, positively skewed, negatively skewed, bimodal) - Summarising the centre and spread of data sets using mean, median and range - Comparing some aspects of the organised and summarised data (ie some of tables, graphs, means, medians and ranges), rather than just relying on one aspect, and deciding whether there is a significant difference between the data sets and	Students compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers, by using comprehensive evidence and providing detailed reasons when comparing more than two data sets. For example, students can: - Compare more than two data sets using comprehensive evidence by: - Choosing and constructing appropriate, easily comparable graphs aligned vertically or horizontally (eg histograms OR divided bar graphs with grouped data and relative frequencies). - Describing the shapes of the data distributions (ie symmetrical, positively skewed, negatively skewed, bimodal). - Summarising the centre and spread of data sets using mean, median and range - Comparing all aspects of the organised and summarised data (ie all of tables, graphs, means, medians and ranges),	Students compare and analyse the distributions of multiple numerical data sets, choose representations, describe features of these data sets using summary statistics and the shape of distributions, and consider the effect of outliers, by examining the relevant strength of different pieces of evidence and designing further investigations to provide stronger evidence For example, students can: - Demonstrate how different graph types for the same data set reveal different features of the data sets and independently use more than one graph type (eg histograms AND divided bar graphs with grouped data and relative frequencies) in conjunction with means, medians and ranges to draw more detailed conclusions when comparing more than two data sets. - Cite the relative strength of different pieces of evidence when providing valid detailed reasons to support inferences. - Identify and plan for data collection to address weaknesses in data set(s) to lead to better substantiated conclusions. - Demonstrate how the same data summary can match multiple data sets of different sizes and different

		whether one set of data is more predictable (ie has less spread) than the other. – Drawing valid conclusions and providing some reasons. • Demonstrate the impact of the shape of the data distribution on the relative positions of the mean and median by changing some data points (especially outliers) and showing how the relative positions of the mean and median change. Use this knowledge to decide whether more emphasis should be placed on the mean or median or both or neither as a 'typical' value when analysing specific data sets by considering the shape of the data.	rather than just relying on one aspect, and deciding whether there is a significant difference between the data sets and whether one set of data is more predictable (ie has less spread) than the other. – Drawing valid conclusions and providing detailed reasons.	distributions to justify why the data summary alone is insufficient when drawing reliable conclusions. (Eg show how different sized class sets of data with different distributions can result in the same values for mean, median and range).
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Students explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Statistics	With support, they explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view. For example, students can: • Choose whether the data collected in supplied real-world scenarios is primary data or secondary data. • Decide whether the primary data collected in supplied real-world scenarios is fair/unbiased or unfair/biased and give at least one valid reason for their decision. • Name at least three reliable sources of secondary data from the internet	They explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view. For example, students can: • Explain the difference between primary and secondary data and the advantages each type, (eg I know that I can trust primary data because I have collected it myself, secondary data has already been collected by someone else so it is available more quickly) and decide whether to use primary or secondary data to answer particular questions, giving valid reasons why. • Evaluate a sampling method designed by themselves or someone	They explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view by checking the reliability of secondary data sources and showing how to modify data presentations to make them fair and unbiased. For example, students can: • Explain why a usually reliable secondary source of data might have specific data sets that are not trustworthy (eg a data set is missing because a survey wasn't done due to COVID; it is hard to collect accurate data in some circumstances, eg Census data may have been collected twice	They explain how sampling techniques and representation can be used to support or question conclusions or to promote a point of view by checking reliable secondary data sources to see if they are truly independent or whether they are actually using the same source. For example, students can: • Search two (or more) different sources of secondary data on the internet (that are reporting on the same issue) to find out whether they are relying on the same source or whether they are truly independent sources. Describe the evidence that confirms whether the data sources are independent or not and describe the attempted

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	and explain why they think each source is reliable. Compare two different supplied representations of the same data and say which one is fair and which one is misleading giving at least one valid reason why.	else to collect primary data, identify factors which could lead to unreliable and unfair data, (eg sample is too small, sample is not random, only one sample was taken instead of comparing results from multiple random samples of adequate size) and improve the sampling method so that it produces fair and reliable data. - Evaluate whether secondary data is reliable by naming the source of the data and commenting on whether they think the source can be trusted, giving valid reasons why. - Demonstrate how different representations of a graph can give different messages, (eg axis scales not starting from zero, axis scales not going up in equal increments) by constructing one accurate and one misleading graph for the same set of data and explaining the impressions conveyed by each graph. - Evaluate strengths, weaknesses and possible sources of bias in data representations from the internet and judge whether the data representation is fair overall, giving valid reasons.	for some people (in the house where they normally live and in the house where they were staying on census night); a worker in a data collection agency has produced a data set without following normal protocols for ensuring data reliability; a worker who was tired at the end of the day accidentally posted the wrong data set), and explains the steps they would take to try and ensure reliability (eg comparing data sets from different and usually reliable sources). - Search a source of secondary data on the internet to find out how the data was collected and comment on how easy it is to identify the data collection method and then comment on how much trust they would put in the data, giving valid reasons. - Find an unfair, misleading or biased data representation and change the representation so that it accurately portrays the data. Explain and justify the need for each modification to the representation.	confirmation process if independence of data sets cannot be confirmed. Comment on how much trust they would put in each data set giving valid reasons including citing strengths and weaknesses.

Probability

Students determine sets of outcomes for compound events and represent these in various ways.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Probability	With support, students determine sets of outcomes for compound events and represent these in various ways. For example, students can: - Follow a process to construct a tree diagram for a compound event consisting of two single events with and without replacement. - Follow a process to construct a two-way table for a compound event consisting of two single events with and without replacement. - Follow a process to construct an organised list of all possibilities for a compound event consisting of two single events with and without replacement. - Follow a process to construct a Venn diagram for a given set of information.	Students determine sets of outcomes for compound events and represent these in various ways. For example, students can: - Use tree diagrams, two-way tables and organised lists to determine all possible outcomes for compound events comprised of two single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of two single events). - Use Venn diagrams with up to three sets to analyse the possible outcomes for compound events (eg construct a Venn diagram to show the numbers of students participating in Tennis, Football or Cross Country and combinations of activities).	Students determine sets of outcomes for compound events and represent these in various ways by analysing compound events which consist of combinations of three single events. For example, students can: Use tree diagrams or organised lists to determine all possible outcomes for compound events comprised of three single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three single events).	Students compare sets of outcomes for different compound events by fluently converting between different representations of compound events which consist of combinations of three (or more) single events. For example, students can: Fluently convert between different representations (ie re-write tree diagrams as organised lists of all possibilities and re-write organised lists of all possibilities as tree diagrams) to compare outcomes for compound events in two different real-world situations (eg rolling two dice and picking a numbered card versus flicking three spinners with numbers on them). The compound events are comprised of three (or more) single events, with and without replacement. (Single events could be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three (or more) single events).

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard																
Probability	With support, they assign probabilities to the outcomes of compound events. For example, students can: - Follow a process to construct a tree diagram, two-way table or organised list to analyse compound events consisting of two single events and hence determine probabilities as fractions then decimals and percentages. - Follow a process to construct a Venn diagram for a given set of information and hence determine probabilities as fractions, decimals and percentages.	They assign probabilities to the outcomes of compound events. For example, students can: - Express probabilities for compound events comprised of two single events, with and without replacement, as fractions, decimals and percentages. (Probabilities as fractions, decimals and percentages can be determined from theoretical analysis (tree diagrams, two-way tables, organised lists or Venn diagrams) or from data in long-run experiments where relative frequency is used (ie fractions, decimals and percentages) instead of actual frequency). - Demonstrate how the words “and” and “or” can affect probabilities for compound events comprised of two single events, with and without replacement. For example, if I spin both of these spinners at the same time  - the list of all possible outcomes is (A stands for Aqua) <table><tr><td>• YY</td><td>• BY</td><td>• RY</td><td>• AY</td></tr><tr><td>• YB</td><td>• BB</td><td>• RB</td><td>• AB</td></tr><tr><td>• YR</td><td>• BR</td><td>• RR</td><td>• AR</td></tr><tr><td>• YA</td><td>• BA</td><td>• RA</td><td>• AA</td></tr></table>	• YY	• BY	• RY	• AY	• YB	• BB	• RB	• AB	• YR	• BR	• RR	• AR	• YA	• BA	• RA	• AA	They assign probabilities to the outcomes of compound events by analysing compound events which consist of combinations of three single events. For example, students can: Express probabilities for compound events comprised of three single events, with and without replacement, as fractions, decimals and percentages. (Probabilities as fractions, decimals and percentages can be determined from theoretical analysis (tree diagrams, organised lists or Venn diagrams) or from data in long-run experiments where relative frequency is used instead of actual frequency).	They compare probabilities for different compound events by converting between different representations of different compound events which consist of combinations of three (or more) single events. For example, students can: Fluently convert between different representations (ie re-write tree diagrams as organised lists of all possibilities and re-write organised lists of all possibilities as tree diagrams) to compare probabilities for compound events in two different real-world situations (eg what is the probability of getting 1, 2 and 3 when rolling two dice and picking a numbered card versus flicking three spinners with numbers on them). The compound events are comprised of three (or more) single events, with and without replacement. (Single events could include: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three single events).
• YY	• BY	• RY	• AY																	
• YB	• BB	• RB	• AB																	
• YR	• BR	• RR	• AR																	
• YA	• BA	• RA	• AA																	

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
		• So: – the probability of getting yellow and blue is $\frac{2}{16}$ which is 12.5% – the probability of getting yellow or blue is $\frac{12}{16}$ which is 75% if the word “or” is being used as “inclusive or”, ie yellow or blue or both – the probability of getting yellow or blue could be $\frac{10}{16}$ which is 62.5% if we are using “exclusive or”, ie yellow or blue but not both. • Use Venn diagrams with up to three circles to determine probabilities for compound events (eg construct a Venn diagram to show the numbers of students participating in Tennis, Football or Cross Country and combinations of activities then determine the probability that a student participates in Tennis and Football but not Cross Country).		

Students design and conduct experiments or simulations for combined events using digital tools.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
Probability	With support, students design and conduct experiments or simulations for combined events using digital tools. For example, students can: • Follow a process to create a digital experiment or simulation for compound events comprised of two single events, with and without replacement. (Single events could	Students design and conduct experiments or simulations for combined events using digital tools. For example, students can: • Use digital tools to design and conduct experiments or simulations for compound events comprised of two single events, with and without replacement. (Single events could	Students design and conduct experiments or simulations for combined events using digital tools for compound events which consist of combinations of three single events. For example, students can: • Use digital tools to design and conduct experiments or	Students design and conduct experiments or simulations for combined events using digital tools for compound events which consist of combinations of three (or more) single events and demonstrate that the same simulation can accurately represent different real-world scenarios.

Strand	Approaching the Standard	At Standard	Above Standard	Well Above Standard
	be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of two single events). Follow a process given by the teacher to check that the results from the digital experiment or simulation are in reasonable agreement with the results that are expected from theoretical analysis (tree-diagram, two-way table or organised list).	be: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of two single events). Compare the results from the digital experiment or simulation with the results that are expected from theoretical analysis (tree-diagram, two-way table or organised list).	simulations for compound events comprised of three single events, with and without replacement. (Single include: tossing a coin, rolling a dice, picking a card, using a spinner, picking a coloured ball out of a container and the compound events could be any combination of three single events). Compare the results from the digital experiment or simulation with the results that are expected from theoretical analysis (tree-diagram or organised list).	For example, students can: Understand that different real-world situations can be equivalent from a probability point of view. Demonstrate this understanding by showing that the same digital simulation generates results that match the theoretical probabilities of two or more different scenarios where each scenario uses different objects and symbol systems (eg one scenario uses numbers on dice, another uses letters on cards and a third scenario uses colours on spinners). The scenarios being compared must be compound events comprised of three (or more) single events, with and without replacement.